

Ahmed Nasser Hemdan

Software Engineer

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Summary

Fourth-year Computer Science student with a solid foundation in software engineering and core computer science concepts. Experienced in building practical applications and solving real-world problems, with a strong focus on writing clean, efficient, and maintainable code. Continuously learning and exploring new technologies, with an interest in developing scalable systems and data-driven applications.

Education

Bachelor of Computer Science, *Cairo University*

Oct 2022 – Jun 2026

Grade: Very Good

Relevant Courses: Machine Learning, Statistics and Probability, Linear Algebra, Calculus, Soft Computing, Algorithms, Data Structures, Discrete Mathematics, Operating Systems, Software Engineering.

Professional Experience

Machine Learning Ambassador

GTC, Sep 2025 – Oct 2025

- Selected among the top candidates from 2,000 applicants to serve as a Machine Learning Ambassador.
- Supported community initiatives by helping peers understand core ML concepts and applications.
- Actively engaged in technical discussions covering core ML topics and data analysis.

iOS Internship

Banque Misr, Aug 2024 – Sep 2024

- Integrated RESTful APIs to enable reliable client-server communication and data-driven application behavior.
- Developed and maintained production-level mobile features within an agile development environment.
- Improved application usability and performance through iterative development and optimization.

Certifications

- **Machine Learning Specialization** – DeepLearning.AI & Stanford University
Completed a hands-on specialization covering supervised and unsupervised learning, neural networks, and ensemble methods with practical implementations.
- **Forward Program – Core Skills Level** – McKinsey & Company
Selective program focused on problem-solving, leadership, and professional communication.

Projects

DocuMind – RAG System

- Built a Retrieval-Augmented Generation (RAG) system that ingests PDF documents and generates AI-powered answers using semantic search
- Implemented PDF processing pipeline with automatic text chunking and vector embeddings using SentenceTransformers
- Integrated Qdrant vector database for efficient similarity search and semantic document retrieval
- Developed responsive Streamlit UI with FastAPI REST backend for seamless user interaction
- **Technologies:** Python, FastAPI, Streamlit, Inngest, Qdrant, SentenceTransformers, Groq API, LLaMA

Tweet-Emotion-Classification – Tweet Emotion Classification using CNN and Word Embeddings

- Developed and compared two CNN-based models using pretrained GloVe embeddings and custom-trained Word2Vec embeddings for emotion classification.
- Implemented text preprocessing, tokenization, sequence padding, embedding matrix construction, and model evaluation using TensorFlow/Keras and Gensim.
- Achieved **93.23% test accuracy** with the CNN + Word2Vec model, outperforming the pretrained GloVe-based model by **1.85 percentage points**.
- **Technologies:** Python, TensorFlow, Keras, Gensim, NLP, CNN, Word2Vec, GloVe, NumPy, Pandas

NFG-Library – Soft computing and intelligent systems library

- Designed and implemented a modular Java library for soft computing techniques, including Neural Networks, Genetic Algorithms, and Fuzzy Logic, with clean and extensible APIs.
- Applied genetic algorithms to optimization problems such as job scheduling, demonstrating solution search, fitness evaluation, and evolutionary operators.
- Developed fuzzy logic controllers for real-world scenarios (e.g., automatic window-blind control), translating linguistic rules into computational inference systems.
- **Technologies:** Java, Object-Oriented Design, Modular Architecture.

BazzarBook – E-commerce platform for book sales

- Designed and developed a multi-user e-commerce platform for managing book sales, orders, and user accounts.
- Implemented role-based access control to support different user roles (e.g., customers and administrators) with secure authorization logic.
- Integrated Stripe payment processing and order management workflows, handling validation, transaction states, and business rules.
- Designed and managed relational data models to support products, users, orders, and transactions.
- **Technologies:** C#, ASP.NET MVC, SQL Server.

Technical Skills

- **Software Engineering Fundamentals:** Object-Oriented Programming, SOLID Principles, Design Patterns
- **Programming Languages:** Python, Java, C++, C
- **ML & Data Libraries:** scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
- **Machine Learning:** Supervised and Unsupervised Learning, Model Training and Evaluation, Feature Engineering, Exploratory Data Analysis (EDA)
- **Databases:** PostgreSQL, MySQL, SQLAlchemy ORM, Alembic
- **Backend:** FastAPI, RESTful APIs, Authentication/Authorization (JWT)
- **Tools:** Git, Linux, Jupyter Notebook

Languages

- Arabic
- English